

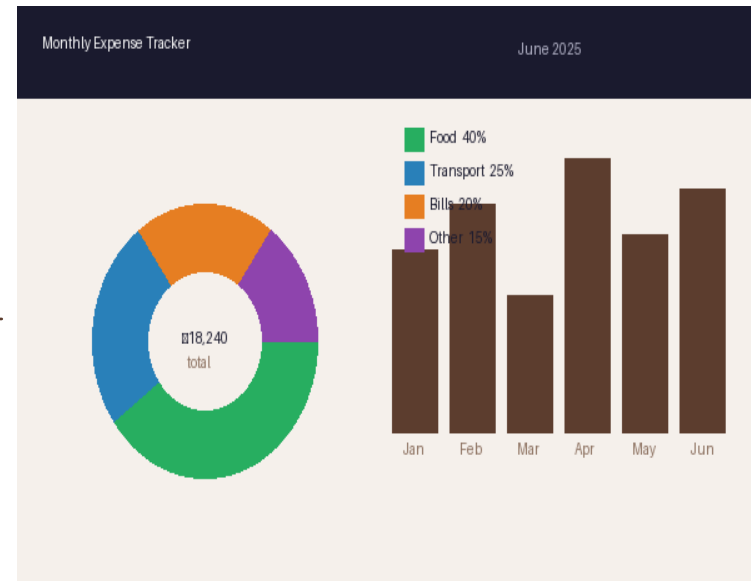
Monthly Expense Tracker

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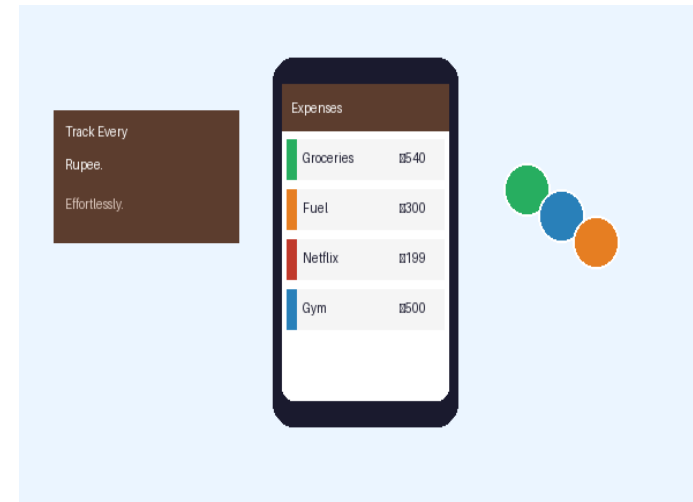
Department of Computer Science

Federal Institute of Science and Technology



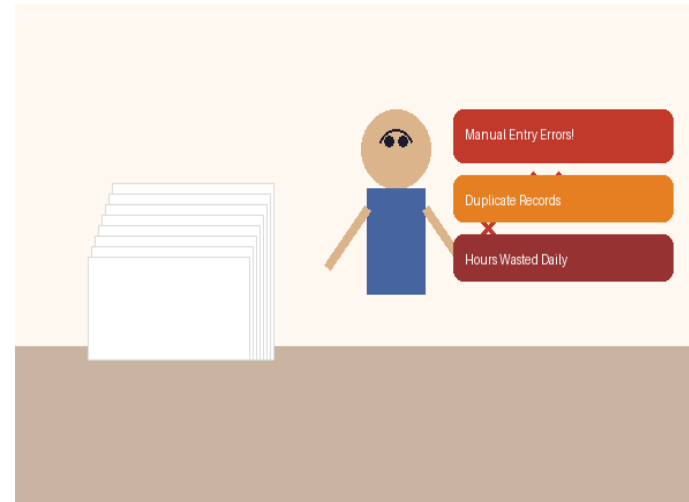
Introduction

- A monthly expense tracker records, categorises, and summarises personal or organisational spending over a calendar month.
- Efficient expense management is critical for budgeting, financial discipline, and meeting savings goals.
- Manual tracking through spreadsheets or paper registers is error-prone, time-consuming, and difficult to analyse at scale.

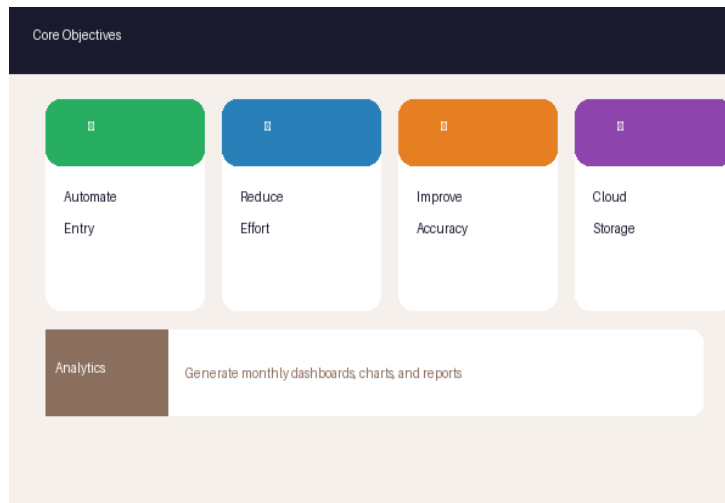


Problem Statement

- Manual data entry of daily expenses is tedious and leads to missed or duplicate records.
- Human errors cause incorrect totals and unreliable monthly summaries.
- High volume of transactions across multiple categories overloads users — insights are delayed.
- There is an urgent need for an accurate, automated expense tracking and reporting solution.



Objectives



Automate Entry

Automatically capture expenses via receipt scanning + OCR.

Reduce Effort

Minimise manual intervention and speed up logging time.

Improve Accuracy

Use validation rules & category-matching to reduce errors.

Cloud Storage & Analytics

Persist structured data in MySQL; generate dashboards for finance.

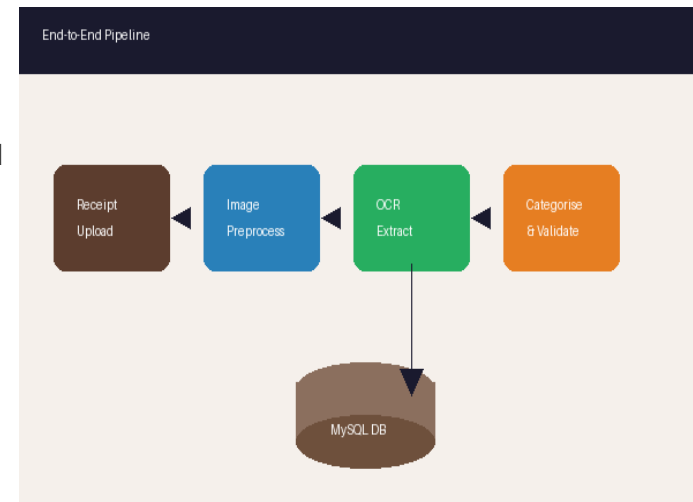
Existing System vs Proposed System

Aspect	Manual Process	Manual vs Automated ⚡ At a Glance	
Speed	Hours per week	Speed Manual: Hours/week	Speed Auto: Minutes/day
Accuracy	Prone to human error	Accuracy Manual: ~70%	Accuracy Auto: ~95%+
Scalability	Limited by discipline	Scalability Manual: Limited	Scalability Auto: Unlimited
Traceability	Manual logs, inconsistent	Cost Manual: High	Cost Auto: Low
Automation delivers: consistent data, faster insights, lower operational effort.			

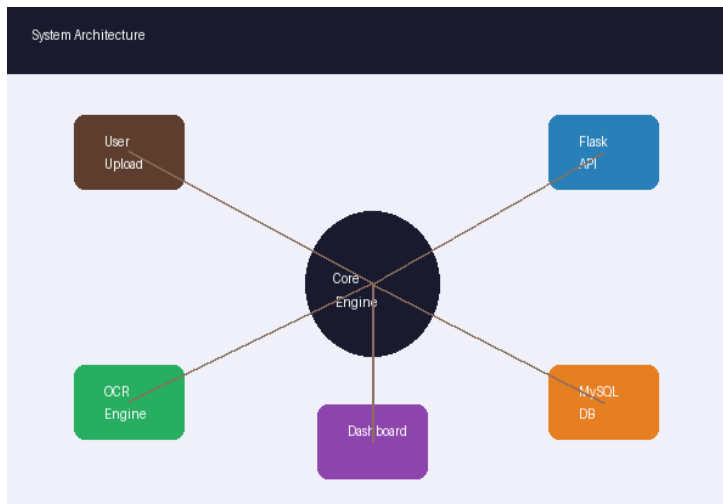
Proposed Solution — Overview

An end-to-end pipeline that accepts uploaded receipts or manual entries, preprocesses images, applies OCR/ML extraction, categorises transactions automatically, stores structured records in MySQL, and exposes dashboards for monthly analytics and reports.

- Accepts JPG, PNG, and PDF receipt uploads.
- Auto-categorises: Food, Transport, Bills, Shopping, Health, Other.
- Monthly summary with charts and downloadable reports.
- Human-in-loop fallback for low-confidence extractions.



System Architecture



User Upload

Web interface or mobile app for receipt photo upload or manual entry.

Image Processing

Preprocessing steps using OpenCV to improve OCR quality.

OCR Engine

Tesseract for text OCR or Donut for receipt document understanding.

Storage & Dashboard

MySQL stores records; Bootstrap UI presents monthly analytics.

Components communicate via Flask REST APIs. Validation supports human-in-loop review for low-confidence fields.

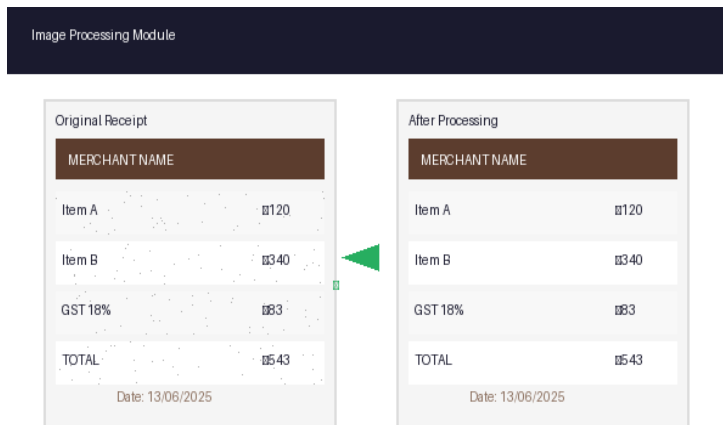
Workflow Diagram

- Upload — User submits a receipt image or PDF via web/mobile.
- Preprocess — OpenCV grayscale, denoise, deskew, threshold.
- OCR — Tesseract / Donut extracts raw text from the image.
- Categorise — NLP model assigns expense category automatically.
- Validate — Confidence check; low-confidence fields queued for review.

Each step includes confidence scoring to maintain accuracy.



Image Processing Module



- Grayscale conversion — simplifies receipt data for OCR.
- Noise removal — median & bilateral filters to reduce artifacts.
- Thresholding — adaptive/Otsu for clearer text-background separation.
- Skew correction — detect and rotate to align text baseline.

Benefits: improves OCR accuracy, reduces false positives, normalises diverse receipt and bill layouts.

OCR & Data Extraction

OCR converts receipt image pixels to text. We combine Tesseract with Hugging Face Donut to handle structured and semi-structured receipts.

- Extraction targets: Merchant, Date, Category, Item Description, Total Amount, GST Number.
- Techniques: pattern matching, layout-aware ML, and confidence thresholds.
- Fallback: human review for fields below confidence threshold.

The diagram illustrates the OCR & Data Extraction process. On the left, a receipt titled "SuperMart Receipt" is shown with fields for Merchant, Date, Items, GST No., and Total. An arrow points from the receipt to a box on the right labeled "Extracted JSON", which contains a JSON object representing the extracted data.

SuperMart Receipt	
Merchant	SuperMart
Date	13/06/2025
Items	4 items
GST No.	27ABCDE1234F
Total	₹1,240

```
Extracted JSON
{
  'merchant': 'SuperMart',
  'date': '2025-06-13',
  'category': 'Grocery',
  'items': 4,
  'gst': '27ABCDE1234F',
  'total': 1240,
  'confidence': 0.96
}
```